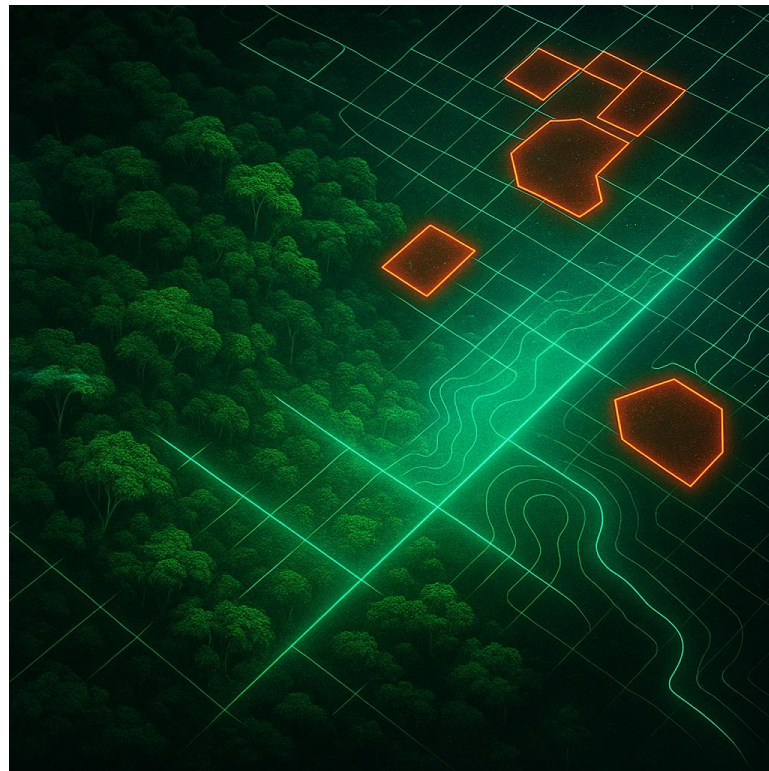


Jungle Anomaly Finder

NDVI Satellite Explorer

AI-guided exploration of hidden Amazonian ruins

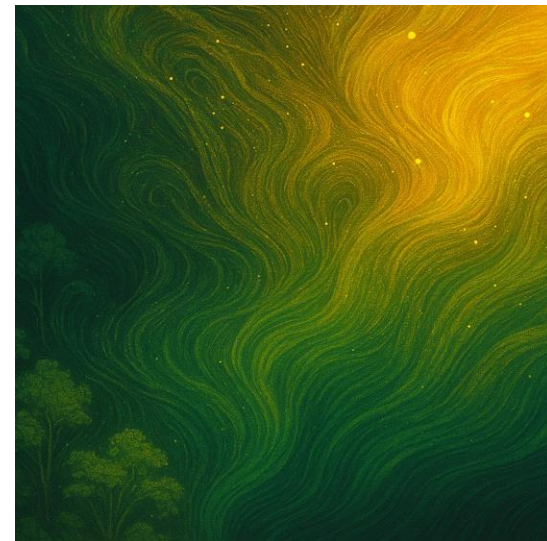
From Zero to Z



July 26, 2025

Introduction & Motivation

- Inspired by the Pororoca tidal bore, symbolising unstoppable exploration through data and intuition.
- Solo entry supported by family, friends and AI—transforming curiosity into discovery.
- Aims to uncover hidden Amazonian archaeological sites using NDVI anomalies, place names and canine intuition.
- Target audience: scientists, archaeologists, local researchers and citizen explorers seeking to challenge traditional satellite analyses.



Problem & Unique Approach

- Conventional satellite analysis misses subtle archaeological signatures in dense rainforest.
- Dense vegetation and hydrology obscure ancient mounds and earthworks.
- Traditional methods overlook ancestral knowledge and animal intuition.



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**NDVI anomaly detection**
Uses time-series NDVI to highlight anomalous vegetation patterns.

**Toponymic & legend analysis**
Mining indigenous place names and legends to reveal hidden clues.

**Canine intuition integration**
Incorporates dog hesitation logs as sensors for unseen anomalies.

**Automated & modular**
Plug-and-play pipeline that is region-agnostic and [open-source](#).

patterns.

Hidden clues.

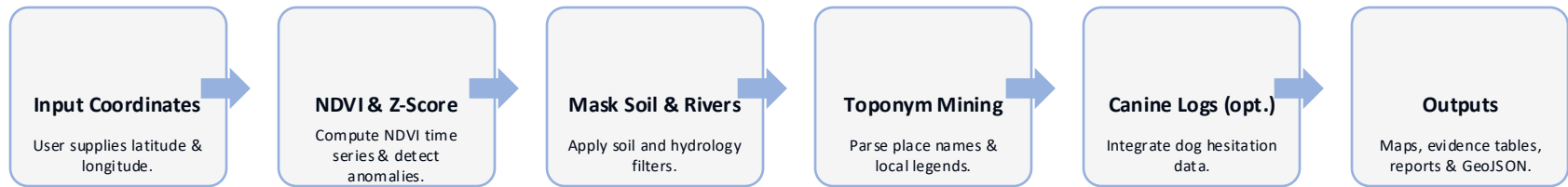
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Automated Pipeline



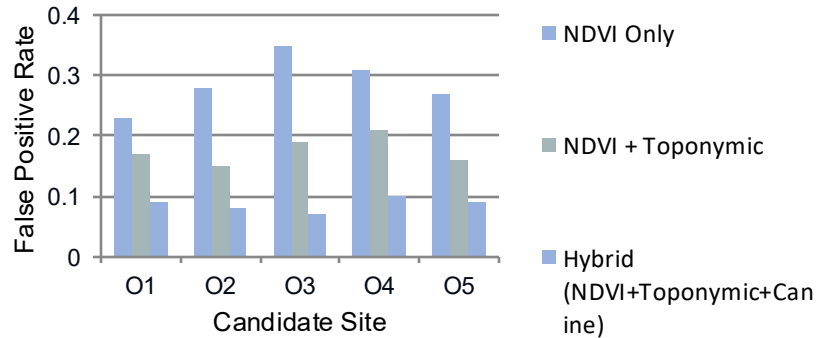
Candidate Sites & AI Scores

Site	NDVI Peak	Z-Score	Anomaly Description	AI Score
O1	0.84	1.7	Subtle vegetation anomaly.	0.10
O2	0.87	1.9	Minor vegetation shift.	0.08
O3	0.89	2.1	Polygonal anomaly hinting at human design.	0.72
O4	0.82	1.5	Diffuse zone with faint patterns.	0.09
O5	0.86	2.0	Linear anomaly near riverbank.	0.33

Five candidate sites were detected via NDVI anomaly analysis. O3 exhibits the strongest signal with a high AI ruin likelihood, while O1–O5 show varying degrees of subtle anomalies.

Performance Analysis

False Positive Rate Reduction



- The hybrid approach dramatically reduces false positives across all sites.
- Combining NDVI with toponymic and canine signals boosts precision and highlights O3 as the most promising site.
- Results emphasise that integrating indigenous knowledge and animal intuition adds measurable value.

Tools & Reproducibility

Category	Tools
Satellite Analysis	Google Earth Engine, Python, Colab
AI & Automation	ChatGPT, Codex, Open Interpreter, Markdown Reports
Collaboration	GitHub + MCP (Model Context Protocol)

- 1. Clone the repository & install dependencies.
- 2. Authenticate Earth Engine (``earthengine authenticate``).
- 3. Run the pipeline: ``python run_pipeline.py --site <SITE_ID>``.
- Outputs: ``results.json``, ``report.md`` and GeoJSON/visual files.

Future Prospects & Impact

- Conduct soil analyses near sacred & restricted zones; focus on adjacent areas to respect cultural boundaries.
- Quantify gains from toponymic and intuition signals versus baseline methods.
- Develop SaaS/API for citizen explorers; adapt methodology to other regions worldwide.
- Field validation requires collaboration with local communities and inclusion of canine companions.
- This open, modular pipeline empowers archaeologists and citizen scientists alike.
- By integrating indigenous wisdom and animal intuition, we honour the past while innovating the future.
- The journey continues: explore boldly, trust intuition, and dance before the gate.

Conclusion & Invitation

- By fusing satellite analytics, ancestral toponyms and canine intuition, our pipeline uncovers hidden stories within the Amazon.
- The hybrid approach reduces false positives and elevates promising sites like O3, showcasing the power of interdisciplinary methods.
- This work invites everyone—scientists, locals, and curious minds—to join the exploration and resonate with the past.
- *Explore boldly. Trust intuition. Embrace AI. Dance before the gate.*

References & Acknowledgments

- PeerJ (2023), Nature (2022), Science (2022) – research on Amazon archaeology & canopy dynamics.
- Google Earth Engine – NDVI & hydrology layers.
- GitHub & Kaggle – Project code & narrative writeup.
- OpenAI Codex & ChatGPT – AI-assisted engineering & narrative writing.

Thank you for joining this exploration journey.

Follow the work under #KGNINJA & #OpenAltoZ.

Z is for the future.

Through this competition, I faced the strongest Kagglers, Ian Adams.
And yet, it was not a battle—but a dialogue.

Together, we opened Z—not as an endpoint, but as the beginning of a
new map.
ZERO to Z.

Let this be a trace for others to follow, or diverge from.
The journey continues.